

Survey Line Planner

Planning Methodology and Equation Reference

Geometry-based MBES line planning - IHO S-44 aware



This document describes the planning workflow, software specification, equations, assumptions, sonar options, beam spacing methodology, recommendation logic, and worked examples for the browser-based Survey Line Planner application.

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Summary of Equation Symbols

The following table summarizes the principal variables and symbols used throughout the manual. It is intended as a quick reference before reading the detailed equations in the planning workflow sections.

Symbol	Meaning	Units	Used in
D_s	Depth source selected by the user from minimum, average, or maximum depth	m	Swath, range, frequency recommendation
D_{min}	Minimum depth entered in the Environment module	m	Planning depth source
D_{avg}	Average depth entered in the Environment module	m	Planning depth source
D_{max}	Maximum depth entered in the Environment module	m	Planning depth source
f	Selected sonar operating frequency	kHz	Sonar recommendation
f_{target}	Target frequency estimated from planning depth	kHz	Frequency recommendation
θ	Requested total operating swath angle	degrees	Swath geometry
θ_{max}	Maximum planning swath angle from the sonar profile	degrees	Swath recommendation
θ_{rec}	Recommended operating swath angle after environmental penalties	degrees	Sonar recommendation
α	Requested half-swath angle, equal to $\theta/2$	degrees	Range limiting
α_r	Maximum half-angle allowed by the selected maximum range	degrees	Range limiting
α_e	Effective half-angle used after range limiting	degrees	Effective swath
θ_e	Effective total swath angle, equal to $2\alpha_e$	degrees	Effective swath
R_s	Slant range required to reach the outer beam	m	Range check
R_{max}	Maximum usable slant range selected in Sonar Settings	m	Range check
W_t	Theoretical swath width for flat-bottom geometry	m	Swath geometry
W_e	Effective swath after range limiting	m	Swath geometry
W_u	Usable planning swath after correction factors	m	Line spacing, density
F_c	Combined swath correction factor	decimal	Usable swath
F_{slope}	Bottom slope correction factor	decimal	Usable swath
F_{sea}	Sea state correction factor	decimal	Usable swath
F_{bottom}	Bottom return correction factor	decimal	Usable swath
F_{IHO}	IHO confidence factor from selected survey order	decimal	Usable swath
S	Mainline spacing	m	Line plan
O	Planned overlap between adjacent swaths	decimal or percent	Line spacing
N_b	Planning beam count or planning soundings per ping	count	Density estimate
K_b	Beam spacing mode coefficient	decimal	Across-track spacing
Δy	Average across-track sounding spacing	m	Density estimate
Δy_{eqd}	Across-track spacing for Equidistant mode	m	Beam spacing
Δy_{eqa}	Across-track spacing for Equiangular mode	m	Beam spacing
Δy_{HD}	Across-track spacing for High Density mode	m	Beam spacing
Δy_i	Spacing between adjacent equiangular beams i and $i + 1$	m	Equiangular beam model

Symbol	Meaning	Units	Used in
θ_i	Individual beam angle for beam i	degrees	Equiangular beam model
V_{kn}	Vessel speed entered by the user	kn	Density, duration
$V_{m/s}$	Vessel speed converted to metres per second	m/s	Density, duration
P	Ping rate	Hz	Along-track spacing, density
P_{min}	Minimum ping rate required to satisfy density at current speed	Hz	Density recommendation
Δx	Along-track sounding spacing	m	Density estimate
Δx_{max}	Maximum allowable along-track spacing for required density	m	Maximum speed
$V_{max,m/s}$	Maximum allowable vessel speed in metres per second	m/s	Maximum speed
$V_{max,kn}$	Maximum allowable vessel speed in knots	kn	Maximum speed
G	Grid cell size	m	Density estimate
A_{cell}	Grid cell area, equal to G^2	m ²	Density estimate
$A_{footprint}$	Nominal sounding footprint area, equal to $\Delta x \Delta y$	m ²	Density estimate
SPC_{est}	Estimated soundings per cell	count / cell	Density result
SPC_{req}	Required soundings per cell	count / cell	Density requirement
L_{area}	Survey area length	m	Survey line length, map preview
W_{area}	Survey area width	m	Survey line count, crossline length
R_{in}	Run-in distance added before each line	m	Line length, duration
R_{out}	Run-out distance added after each line	m	Line length, duration
L_{line}	Average planned survey line length including run-in and run-out	m	survey line length
N_{line}	Number of survey lines	count	Line plan, duration
L_{total}	Total planned mainline length	m or km	Crosslines, duration
X	Crossline requirement as percentage of survey line length	decimal or percent	Crossline planning
$L_{cross,req}$	Required crossline length based on $L_{total}X$	m	Crossline planning
$L_{cross,line}$	Average planned crossline length	m	Crossline planning
N_{cross}	Number of crosslines	count	Crossline planning, duration
S_{cross}	Crossline spacing across the survey area	m	Crossline planning
L_{cross}	Total planned crossline length	m or km	Duration
A_{main}	Survey line azimuth	degrees	Map preview, crossline azimuth
A_{cross}	Crossline azimuth, perpendicular to survey line azimuth	degrees	Crossline planning
T_{run}	Total line running time for survey lines and crosslines	h	Duration
T_{turn}	Turn time per line or line-change allowance	min / turn	Duration
$T_{turn,total}$	Total accumulated turn time	h	Duration
$T_{subtotal}$	Runtime plus turn time before operational allowance	h	Duration
A_{ops}	Operational allowance added to subtotal time	decimal or percent	Duration
T_{total}	Final estimated survey time including allowance	h	Duration result

1 Document Control

1.1 Purpose of This Manual

This technical manual documents the methodology and software behavior of the Survey Line Planner. It is intended to be used as the master reference for future updates to the browser application and for explaining how the planner generates recommendations, warnings, and survey-line estimates.

1.2 Revision Record

Version	Date	Author	Description
1.0	2026-06-07	AAA	Published planning methodology reference

1.3 Key Technical Changes from the Previous Version

- This is the first complete iteration of the manual.

2 Introduction

The Survey Line Planner is a hydrographic survey planning application designed to assist surveyors in estimating multibeam echosounder (MBES) line spacing, sounding density, vessel speed limitations, survey line requirements, crossline requirements, and survey duration.

The planner is intended for planning and feasibility assessment. It does not replace field validation, hydrographer judgment, manufacturer guidance, project specifications, or quality-control procedures.

3 Purpose and Planning Philosophy

The planner balances three competing planning objectives:

- **Coverage** - sufficient overlap and swath coverage to meet the survey objective.
- **Density** - sufficient soundings per grid cell to satisfy the selected survey requirement.
- **Efficiency** - practical survey duration and realistic acquisition speed.

The planner therefore seeks the largest practical line spacing that still satisfies the required sounding density, selected overlap, sonar capability limits, environmental limitations, and user-defined planning constraints.

4 Software Overview

The application is divided into five major input modules. Each module collects a specific group of planning inputs and passes those values to the recommendation engine and calculation modules.

1. Survey Requirements
2. Environment Conditions
3. Sonar Configurations
4. Vessel and Planning Parameters
5. Survey Area Geometry

4.1 Survey Requirements Module

The Survey Requirements module defines the required survey standard and the minimum density target that the resulting line plan must satisfy.

4.1.1 Purpose

This module establishes the survey compliance target. It sets the survey order, required grid cell size, required soundings per cell, and compliance percentage. The selected IHO order drives the recommended overlap, full seafloor search state, and recommended crossline percentage, while the grid cell size and required soundings per cell are used to evaluate whether the planned line spacing, speed, ping rate, and beam spacing meet the required sounding density.

4.1.2 Inputs

Input	Symbol	Sample value	Planner use
IHO order	–	Order 1a	Selects the planning requirement profile and automatically sets default grid size, required SPC, full seafloor search state, overlap, and crosslines.
Grid cell size	G	1 m	Defines the cell area used for sounding-density calculations.
Required soundings per cell	SPC_{req}	5	Defines the minimum acceptable estimated sounding density.
Compliance percentage	C_{req}	95%	Documents the target confidence or project compliance level. It is displayed as a requirement input and may be used in future QA logic.
Overlap	O	30%	Reduces usable swath to line spacing and increases coverage redundancy.
Crossline percentage	P_{QC}	8%	Defines the recommended QC crossline length as a percentage of total survey line length.

4.1.3 Calculations and Logic Driven by the Module

- Line spacing: $S = W_u(1 - O)$.
- Density pass/fail check: $SPC_{est} \geq SPC_{req}$.
- Recommended crossline length: $L_{QC,req} = L_{total}P_{QC}$.
- Recommendation badges, full seafloor search state, and warning logic.

Sample survey requirement values

For the sample project:

IHO order = Order 1a

$G = 1$ m

$SPC_{req} = 5$

$C_{req} = 95\%$

$O = 30\% = 0.30$

$P_{QC} = 8\% = 0.08$

4.2 Environment Conditions Module

The Environment Conditions module describes the depth and expected survey conditions that affect usable swath, frequency recommendations, range checks, and coverage confidence.

4.2.1 Purpose

This module converts field conditions into planning constraints. Depth controls theoretical swath and slant range. Slope, sea state, and bottom return reduce practical usable swath to account for degraded bottom detection and reduced confidence in outer-beam data.

4.2.2 Inputs

Input	Symbol	Sample value	Planner use
Average depth	D_{avg}	30 m	Used as planning depth when the planning depth source is Average.
Minimum depth	D_{min}	20 m	Used as planning depth when the planning depth source is Minimum. This is the conservative shallow-water choice.
Maximum depth	D_{max}	40 m	Used as planning depth when the planning depth source is Maximum. This is useful for deep-water coverage sensitivity.
Depth source	D_s	Average	Selects the source of the depth calculations whether D_s is taken from minimum, average, or maximum depth.
Sea state	F_{sea}	Calm	Selects the sea-state reduction factor F_{sea} .

Bottom slope	F_{slope}	Flat	Selects the slope reduction factor F_{slope} and contributes to swath-angle recommendations.
Bottom return	F_{bottom}	Good	Selects the bottom-return reduction factor F_{bottom} .

4.2.3 Calculations and Logic Driven by the Module

- Planning depth selection: $D_p \in \{D_{min}, D_{avg}, D_{max}\}$.
- Theoretical swath: $W_t = 2D_p \tan(\theta/2)$.
- Slant range: $R_s = D_p / \cos(\theta/2)$.
- Usable swath reduction: $W_u = W_e F_{slope} F_{sea} F_{bottom}$.
- Frequency and swath-angle recommendation logic.

Sample environment values

For the sample project:

$$\begin{aligned}
 D_{avg} &= 30 \text{ m} \\
 D_{min} &= 20 \text{ m} \\
 D_{max} &= 40 \text{ m} \\
 D_s &= D_{avg} = 30 \text{ m} \\
 F_{sea} &= 1.00 \\
 F_{slope} &= 1.00 \\
 F_{bottom} &= 1.00
 \end{aligned}$$

4.3 Sonar Settings Module

The Sonar Settings module defines the sonar capability profile and the acquisition settings used to estimate swath, sounding density, and recommended operating limits.

4.3.1 Purpose

This module connects the selected sonar model with the planning equations. It provides supported frequencies, planning soundings per ping, maximum swath angle, beam spacing mode, ping rate, and range limits. These values affect both geometric coverage and density.

4.3.2 Inputs

Input	Symbol	Sample value	Planner use
Sonar model	–	Kongsberg EM2040	Loads the planning capability profile, supported frequencies, beam spacing modes, planning range values, and recommended defaults.
Frequency	f	300 kHz	Used to evaluate the selected operating band and recommended range profile.

Swath angle	θ	135°	Controls theoretical swath width and slant range.
Soundings per ping	N_b	512	Used as the across-track sounding count for density estimation.
Ping rate	P	5 Hz	Controls along-track spacing and density at the selected vessel speed.
Maximum range	R_{max}	140 m	Limits the effective outer-beam angle if the requested swath exceeds usable sonar range.
Beam spacing mode	–	Equidistant	Determines which across-track spacing model is used: equidistant or equiangular.

4.3.3 Calculations and Logic Driven by the Module

- Theoretical swath and slant-range validation.
- Effective swath if $R_s > R_{max}$.
- Across-track spacing Δy from beam spacing mode.
- Along-track spacing $\Delta x = V_{m/s}/P$.
- Estimated sounding density $SPC_{est} = G^2/(\Delta x \Delta y)$.
- Minimum ping rate and maximum vessel speed.

Sample sonar values

For the sample project:

Sonar model = Kongsberg EM2040

$f = 300$ kHz

$\theta = 135^\circ$

$N_b = 512$

$P = 5$ Hz

$R_{max} = 140$ m

Beam spacing = Equidistant

4.4 Vessel and Planning Parameters Module

The Vessel and Planning Parameters module represents the operational behaviour of the survey platform and the acquisition strategy selected by the hydrographer.

4.4.1 Purpose

This module converts vessel operations into planning constraints. It determines whether the selected vessel speed, overlap, ping rate, turn time, and operational allowance can realistically satisfy the required sounding density and survey duration objectives.

4.4.2 Inputs

Input	Symbol	Sample value	Planner use
Vessel speed	V_{kn}	6 kn	Converts to metres per second and controls along-track spacing, density checks, maximum speed checks, and acquisition time.
Overlap	O	30%	Reduces usable swath to line spacing and increases redundancy.
Turn time	T_{turn}	12 min/turn	Estimates time required to turn or reposition between survey lines.
Allowance	A_{ops}	15%	Adds contingency time for line changes, manoeuvring, weather interruptions, operational inefficiency, and field delays.
Crossline percentage	P_{QC}	8%	Defines the recommended crossline length as a percentage of total survey line length.

4.4.3 Calculations and Logic Driven by the Module

- Speed conversion: $V_{m/s} = 0.514444V_{kn}$.
- Along-track spacing: $\Delta x = V_{m/s}/P$.
- Line spacing from overlap: $S = W_u(1 - O)$.
- Maximum speed check: $V_{max, kn} = (\Delta x_{max}P)/0.514444$.
- Survey duration including run time, turn time, and operational allowance.

Sample vessel and planning values

For the sample project:

$$\begin{aligned}
 V_{kn} &= 6 \text{ kn} \\
 O &= 30\% = 0.30 \\
 T_{turn} &= 12 \text{ min/turn} \\
 A_{ops} &= 15\% = 0.15 \\
 P_{QC} &= 8\% = 0.08
 \end{aligned}$$

4.5 Survey Area Module

The Survey Area module defines the rectangular planning area used by the schematic line planner. It determines how many survey lines are required, how long each line is, where crosslines are distributed, and how much total acquisition distance is expected.

4.5.1 Purpose

This module converts the user-defined survey area into distances that can be combined with line spacing and vessel speed. It is placed after survey requirements, environment, sonar settings, and vessel planning because the area dimensions do not determine sonar capability; instead, they scale the final line count and duration once spacing has been calculated.

4.5.2 Inputs

Input	Symbol	Sample value	Planner use
Survey length	L_{area}	1500 m	Defines nominal survey line length before run-in and run-out are added.
Survey width	W_{area}	600 m	Defines how many survey lines are required for the calculated line spacing.
Azimuth	A_{main}	90°	Used for schematic orientation and crossline azimuth if enabled.
Run-in distance	R_{in}	50 m	Added before the survey area to allow stabilization and full coverage at the start of each line.
Run-out distance	R_{out}	50 m	Added after the survey area to maintain coverage and line stability beyond the boundary.

4.5.3 Calculations and Logic Driven by the Module

- Number of survey lines: $N_{line} = \lceil W_{area}/S \rceil + 1$.
- Average survey line length: $L_{line} = L_{area} + R_{in} + R_{out}$.
- Total survey line length: $L_{total} = N_{line}L_{line}$.
- Average crossline length: $L_{QC,line} = W_{area} + R_{in} + R_{out}$.
- Number of crosslines: $N_{QC} = \lceil L_{QC,req}/L_{QC,line} \rceil$.
- Crossline spacing: $S_{QC} = L_{area}/(N_{QC} + 1)$.
- Crossline azimuth, when used: $A_{cross} = (A_{main} + 90^\circ) \bmod 360^\circ$.

Sample survey area values

For the sample project:

$$L_{area} = 1500 \text{ m}$$

$$W_{area} = 600 \text{ m}$$

$$A_{main} = 90^\circ$$

$$R_{in} = 50 \text{ m}$$

$$R_{out} = 50 \text{ m}$$

5 User Workflow

The recommended workflow is:

1. Select IHO Order.
2. Review automatically populated grid size, required SPC, recommended overlap, and crosslines.
3. Confirm or modify recommendations.
4. Enter environmental conditions which includes survey depths and survey conditions.

5. Select planning depth source.
6. Select sonar model.
7. Review recommended sonar configurations.
8. Enter vessel speed, turn time, and operational allowance.
9. Enter survey area dimensions and planned run-in/run-out distances.
10. Review plan summary which includes the calculated swath width, density, survey lines, crosslines, and duration results.

The planner recalculates continuously whenever an input changes.

6 Survey Requirement Setup

Survey requirement inputs define the quality target for the plan.

6.1 Inputs

- IHO Order
- Grid Size
- Soundings Per Cell (SPC)
- Compliance Percentage

6.2 Important Interpretation

The IHO order selection makes the planner **IHO-aware**. It does not mean the application certifies compliance with IHO S-44. Final compliance must be determined from actual survey data, field QC, uncertainty analysis, project specifications, and applicable hydrographic authority requirements.

6.3 IHO Order Configuration

The planner uses application defaults associated with each IHO Order. These defaults provide a practical starting point for planning and may be overridden by the user.

Order	Grid Size	SPC	Full Search	Overlap	Recommended Crosslines
Exclusive	0.5 m	10	On	50%	12%
Special	1 m	5	On	30%	10%
Order 1a	2 m	3	On	25%	8%
Order 1b	5 m	2	Off	10%	5%
Order 2	10 m	1	Off	0%	5%

Recommended overlap and crosslines are planner defaults based on common hydrographic practice and application design. They are not prescribed as fixed percentages by IHO S-44.

6.3.1 Exclusive Order

Planner effect: highest redundancy, smallest line spacing, largest number of lines, longest acquisition time.

6.3.2 Special Order

Planner effect: high coverage confidence and conservative survey spacing for critical areas.

6.3.3 Order 1a

Planner effect: balanced hydrographic survey setup for full seafloor search in navigationally relevant areas.

6.3.4 Order 1b

Planner effect: reduced redundancy and wider spacing where feature detection is not the primary objective.

6.3.5 Order 2

Planner effect: widest line spacing and fastest acquisition for general bathymetric mapping.

Full Seafloor Search state does not modify the swath geometry equations. Instead, it reduces the recommended swath angle by 5° , which indirectly reduces the theoretical, effective, and usable swath used for line spacing.

7 Environment Setup

Environmental conditions influence achievable coverage, sounding density, and data quality.

7.1 Inputs

- Average depth
- Minimum depth
- Maximum depth
- Depth source
- Sea state: Calm, Moderate, Rough
- Bottom slope: Flat, Moderate, Steep, Rugged
- Bottom return: Good, Moderate, Weak

7.2 Depth Source Selection

The planner may use minimum, average, or maximum depth as the planning depth. This selected planning depth is used in all swath and density equations.

$$D_s = \begin{cases} D_{min}, & \text{if planning depth source is Minimum} \\ D_{avg}, & \text{if planning depth source is Average} \\ D_{max}, & \text{if planning depth source is Maximum} \end{cases} \quad (1)$$

Sample calculation

Depth source = Average. Therefore:

$$D_s = D_{avg} = 30 \text{ m}$$

8 Sonar Setup

The sonar model defines planning capability profiles. The planner does not model every manufacturer-specific setting; it stores planning-relevant characteristics.

8.1 Sonar Capability Profiles

Model	Frequencies	Beam Count	Max Swath	Planning Ranges
Custom MBES	200/300/400 kHz	512/1024	140°	User defined
Kongsberg EM2040	200/300/400 kHz	512/1024	140°	400: 80 m; 300: 140 m; 200: 250 m
Kongsberg EM2042	200/300/400 kHz	512/1024	140°	400: 90 m; 300: 160 m; 200: 300 m
Kongsberg EM712	40/70/100 kHz	512/1024	140°	100: 500 m; 70: 1000 m; 40: 1500 m
Kongsberg EM124	12 kHz	512/1024	150°	12: 6000 m+
R2Sonic 2024	200/300/400 kHz	512/1024	160°	400: 100 m; 300: 180 m; 200: 300 m
R2Sonic 2026	170/200/300/400 kHz	512/1024	160°	400: 120 m; 300: 220 m; 200: 350 m; 170: 450 m
Norbit iWBMS	200/400/700 kHz	512/1024	140°	700: 60 m; 400: 100 m; 200: 180 m
Teledyne SeaBat T20	200/300/400 kHz	512/1024	140°	400: 80 m; 300: 110 m; 200: 180 m
Teledyne SeaBat T50	200/300/400 kHz	512/1024	140°	400: 100 m; 300: 150 m; 200: 250 m

Planning ranges are application planning assumptions used for feasibility checks. They are not certified manufacturer limits. Final values should be verified with manufacturer documentation, acquisition software, field testing, and actual bottom detection performance.

8.2 Effect of Sonar Model Selection

Changing the sonar model can change:

- frequency recommendation;
- maximum available swath angle;
- maximum range validation;
- density estimate through soundings per ping;
- recommended ping rate and maximum vessel speed;
- line spacing and total survey duration.

8.3 Frequency Recommendation Methodology

The planner selects the highest available frequency capable of achieving the required slant range.

$$f_{rec} = \max \{f \in F_{sonar} : R_{sonar}(f) \geq R_{required}\} \quad (2)$$

Where F_{sonar} is the set of available frequencies for the selected sonar model.

Higher frequencies generally provide smaller acoustic footprints and better feature resolution, but lower range. Lower frequencies provide greater range but lower spatial resolution. The planner therefore chooses the highest frequency that can still satisfy the required range.

8.4 Swath Angle Recommendation Methodology

The recommended swath angle begins with the maximum supported swath angle for the selected sonar and is reduced based on survey risk and environmental conditions.

$$\theta_{rec} = \text{clamp}(\theta_{max} - \Delta\theta_{search} - \Delta\theta_{slope} - \Delta\theta_{sea}, 90^\circ, \theta_{max}) \quad (3)$$

Condition	Angle Penalty
Full Seafloor Search	5°
Moderate Slope	10°
Steep Slope	20°
Rugged Terrain	30°
Calm Sea State	0°
Moderate Sea State	10°
Rough Sea State	25°

8.5 Maximum Range Validation

Maximum range validation checks whether the selected sonar and frequency can support the required outer-beam slant range. If the required range exceeds the sonar planning range, the planner reduces the effective swath angle and warns the user.

8.5.1 Required Slant Range

Required slant range is the distance from the sonar to the outer edge of the selected swath. It increases rapidly as the swath angle becomes wider, especially in deeper water.

$$R_{required} = \frac{D_p}{\cos\left(\frac{\theta}{2}\right)} \quad (4)$$

Sample calculation

For $D_p = 30$ m and $\theta = 140^\circ$:

$$R_{required} = \frac{30}{\cos(70^\circ)} = 87.7 \text{ m}$$

8.5.2 Range Validation

Range validation compares the required slant range against the sonar planning range. When the required range is greater than the available range, the selected swath angle cannot be fully achieved.

$$R_{required} \leq R_{sonar} \quad (5)$$

If this condition is not satisfied, the planner reduces effective swath angle, recommends a lower frequency if available, and warns the user.

8.6 Swath Geometry

Swath geometry estimates the width of seafloor coverage produced by the selected depth and swath angle. These calculations provide the starting point for usable swath, line spacing, density, and survey-line planning.

8.6.1 Theoretical Swath Width

For a flat bottom, the theoretical swath width is calculated from the planning depth and total swath angle. This value provides the geometric basis for the subsequent effective swath, usable swath, line spacing, and density calculations.

$$W_t = 2D_s \tan\left(\frac{\theta}{2}\right) \quad (6)$$

Sample calculation

$$W_t = 2(30) \tan(70^\circ) = 164.8 \text{ m}$$

8.7 Range-Limited Effective Angle

If the selected swath requires a slant range larger than the maximum usable range, the effective half angle is reduced.

$$\alpha = \frac{\theta}{2} \quad (7)$$

$$\alpha_r = \cos^{-1} \left(\frac{D_p}{R_{sonar}} \right) \quad (8)$$

$$\alpha_e = \min(\alpha, \alpha_r) \quad (9)$$

$$\theta_e = 2\alpha_e \quad (10)$$

8.8 Effective Swath Width

$$W_e = 2D_p \tan(\alpha_e) \quad (11)$$

8.9 Environmental Reduction Factors

Category	Condition	Factor
Bottom Return	Good	1.00
	Moderate	0.94
	Weak	0.86

These factors are empirical planning values used to reduce the effective swath to a more conservative usable swath. They are not IHO S-44 requirements and should be treated as planning assumptions.

In this methodology, slope, sea state, and full seafloor search state are already accounted for in the recommended swath-angle selection. To avoid reducing the swath twice, the usable-swath correction is limited to bottom return, which represents confidence in bottom detection after the geometric and range-based swath has been calculated.

The selected bottom-return factor is used as the combined correction factor, F_c . This factor is applied to the effective swath width, W_e , to calculate the usable planning swath, W_u , which is then used for line spacing and sounding-density calculations.

$$F_c = F_{bottom} \quad (12)$$

$$W_u = W_e F_c \quad (13)$$

No IHO confidence factor is applied. IHO Order affects planner defaults and recommendations, not swath reduction.

8.10 Beam Spacing Modes

Beam spacing mode affects the estimated across-track sounding spacing Δy and therefore affects the density estimate.

8.10.1 Equidistant Beam Spacing

Equidistant mode assumes soundings are approximately evenly distributed across the usable swath.

$$\Delta y = \frac{W_u}{N_b - 1} \quad (14)$$

Sample calculation

For $W_u = 164.8$ m and $N_b = 512$:

$$\Delta y = \frac{164.8}{511} = 0.322 \text{ m}$$

8.10.2 Equiangular Beam Spacing

Equiangular mode spaces beams by equal angular interval. Seafloor spacing increases toward the outer beams. The true spacing between adjacent beams at beam angles θ_i and θ_{i+1} is approximately:

$$\Delta\theta = \frac{\theta}{N_b - 1} \quad (15)$$

$$\theta_i = -\frac{\theta}{2} + i\Delta\theta \quad (16)$$

$$y_i = D_p \tan(\theta_i) \quad (17)$$

$$\Delta y_i = y_{i+1} - y_i \quad (18)$$

$$\Delta y_{design} = \max(\Delta y_i) \quad (19)$$

The planner uses Δy_{design} for conservative density planning.

9 Sounding Density Methodology

The sounding density methodology estimates whether the planned acquisition will collect enough soundings within each grid cell. It combines vessel speed, ping rate, across-track beam spacing, and grid cell size to calculate the expected soundings per cell.

9.1 Vessel Speed Conversion

Vessel speed is converted from knots to metres per second so it can be used consistently with metric spacing and distance calculations. This converted speed is used for along-track spacing, maximum speed, and survey duration estimates.

$$V_{m/s} = V_{kn} \times 0.514444 \quad (20)$$

9.2 Along-Track Spacing

Along-track spacing is the distance travelled by the vessel between consecutive pings. Higher vessel speed increases this spacing, while a higher ping rate reduces it and increases sounding density.

$$\Delta x = \frac{V_{m/s}}{PR} \quad (21)$$

9.3 Grid Cell Area

Grid cell area is calculated from the selected grid cell size. This area represents the target cell used to evaluate whether enough soundings are expected within each cell.

$$A_{cell} = G^2 \quad (22)$$

9.4 Expected Soundings Per Cell

Expected soundings per cell is estimated by comparing the grid cell area with the nominal sounding footprint area. If the estimated value is greater than or equal to the required soundings per cell, the density requirement is considered satisfied.

$$SPC_{est} = \frac{A_{cell}}{\Delta x \Delta y} \quad (23)$$

Sample calculation

For $G = 2$ m, $\Delta x = 0.617$ m, and $\Delta y = 0.322$ m:

$$SPC_{est} = \frac{4}{0.617(0.322)} = 20.1$$

The requirement $SPC_{req} = 3$ is satisfied.

9.5 Ping Rate Recommendation

Solving the density equation for allowable along-track spacing gives:

$$\Delta x_{max} = \frac{A_{cell}}{SPC_{req} \Delta y} \quad (24)$$

$$PR_{min} = \frac{V_{m/s}}{\Delta x_{max}} \quad (25)$$

9.6 Maximum Vessel Speed

$$V_{max,m/s} = \Delta x_{max} PR \quad (26)$$

$$V_{max,kn} = \frac{V_{max,m/s}}{0.514444} \quad (27)$$

10 Line Calculations

The line plan is calculated by converting the usable swath into a practical survey-line layout. The planner applies the selected overlap to determine line spacing, then compares that spacing against the survey area width to estimate the required number of survey lines. The survey area length, run-in, and run-out distances are then used to calculate the average line length and total planned survey-line distance.

10.1 Line Spacing

Line spacing is the distance between adjacent survey lines. It is calculated by reducing the usable swath by the planned overlap, where higher overlap produces closer line spacing and greater redundancy.

$$S = W_u(1 - O) \quad (28)$$

10.2 Number of Survey lines

The number of survey lines is calculated from the survey area width and the planned line spacing. The ceiling function rounds partial coverage up to a full line, and one additional line is added for boundary coverage.

$$N_{line} = \left\lceil \frac{W_{area}}{S} \right\rceil + 1 \quad (29)$$

10.3 Line Length and Total Survey line Length

The average survey-line length includes the survey area length plus run-in and run-out distances. The total survey-line length is then calculated by multiplying the average line length by the number of planned survey lines.

$$L_{line} = L_{area} + R_{in} + R_{out} \quad (30)$$

$$L_{total} = N_{line}L_{line} \quad (31)$$

10.4 Vessel and Area Interaction

The line planning equations are controlled by both the Vessel and Planning module and the Survey Area Geometry module. The vessel module contributes speed, overlap, turn time, and operational allowance. The area module contributes length, width, run-in, and run-out. Together, they determine the operational scale of the plan.

$$S = W_u(1 - O) \quad (32)$$

$$N_{line} = \left\lceil \frac{W_{area}}{S} \right\rceil + 1 \quad (33)$$

$$L_{line} = L_{area} + R_{in} + R_{out} \quad (34)$$

$$L_{total} = N_{line}L_{line} \quad (35)$$

Sample vessel-area interaction

Using $W_u = 137.6$ m and $O = 0.30$:

$$S = 137.6(1 - 0.30) = 96.3 \text{ m}$$

$$N_{line} = \left\lceil \frac{600}{96.3} \right\rceil + 1 = 8$$

$$L_{line} = 1500 + 50 + 50 = 1600 \text{ m}$$

$$L_{total} = 8(1600) = 12800 \text{ m}$$

10.5 Recommended Crossline Calculation

Recommended crosslines provide independent checks across survey lines. They are used to assess depth consistency, systematic bias, sound velocity effects, line-to-line agreement, motion or attitude errors, and coverage consistency.

$$L_{QC,req} = L_{total}P_{QC} \quad (36)$$

$$L_{QC,line} = W_{area} + R_{in} + R_{out} \quad (37)$$

$$N_{QC} = \left\lceil \frac{L_{QC,req}}{L_{QC,line}} \right\rceil \quad (38)$$

$$S_{QC} = \frac{L_{area}}{N_{QC} + 1} \quad (39)$$

$$L_{QC} = N_{QC}L_{QC,line} \quad (40)$$

11 Survey Duration Estimation

The survey duration model converts the planned line layout into an operational time estimate. The total running time, T_{run} , is calculated from the combined survey line length and QC crossline length divided by the vessel speed in metres per second. The turn-time component, $T_{turn,total}$, accounts for the estimated time required to turn or reposition between each survey line and crossline. The subtotal is then increased by the operational allowance, A_{ops} , to produce the final estimated survey duration, T_{total} .

$$T_{run} = \frac{L_{total} + L_{QC}}{V_{m/s} \times 3600} \quad (41)$$

$$T_{turn,total} = \frac{(N_{line} + N_{QC})T_{turn}}{60} \quad (42)$$

$$T_{subtotal} = T_{run} + T_{turn,total} \quad (43)$$

$$T_{total} = T_{subtotal}(1 + A_{ops}) \quad (44)$$

12 Recommendation Engine

Each recommended value should be traceable to a planning basis.

Recommended Value	Basis
Grid Size	Selected IHO Order planning default
Required SPC	Selected IHO Order planning default
Full Seafloor Search	Selected IHO Order planning default or user toggle
Frequency	Highest available frequency satisfying required slant range
Swath Angle	Sonar maximum minus full-search and condition-based penalties
Maximum Range	Sonar-frequency planning range
Overlap	Selected IHO Order and full-search objective
Planning Soundings per Ping	Sonar profile and user selection
Ping Rate	Required SPC, vessel speed, and beam spacing
Recommended QC Crosslines	Percentage of total survey line length

13 Warning Logic

The planner generates warnings when density, range, speed, overlap, or environmental constraints are not satisfied.

13.1 Density Warning

This warning is triggered when the estimated soundings per cell is lower than the required soundings per cell. It indicates that the plan may need a slower vessel speed, higher ping rate, narrower spacing, or different beam spacing mode.

$$SPC_{est} < SPC_{req} \quad (45)$$

13.2 Range Warning

This warning is triggered when the required outer-beam slant range exceeds the selected sonar planning range. It indicates that the chosen swath angle may be too wide for the depth and sonar range, causing the effective swath to be reduced.

$$R_{required} > R_{sonar} \quad (46)$$

13.3 Speed Warning

This warning is triggered when the current vessel speed exceeds the calculated maximum speed for the required sounding density. It indicates that the vessel should slow down or the acquisition settings should be adjusted to maintain the required density.

$$V_{current} > V_{max} \quad (47)$$

14 Worked Examples

14.1 EM2040 Shallow-Water Planning

14.1.1 Input Values

Parameter	Symbol	Value
IHO Order	-	Order 1a
Grid Size	G	2 m
Required SPC	SPC_{req}	3
Full Seafloor Search	-	On
Sonar Model	-	Kongsberg EM2040
Frequency	f	300 kHz
Soundings per Ping	N_b	512
Beam Mode	-	Equidistant
Planning Depth	D_p	30 m
Swath Angle	θ	140°
Sonar Range	R_{sonar}	140 m
Vessel Speed	V_{kn}	6 kn
Ping Rate	PR	5 Hz
Overlap	O	30%
Survey Length	L_{area}	1500 m
Survey Width	W_{area}	600 m
Run-in / Run-out	R_{in}, R_{out}	50 m / 50 m
Recommended QC Crosslines	P_{QC}	8%
Turn Time	T_{turn}	12 min
Operational Allowance	A_{ops}	15%

14.1.2 Calculation Summary

Worked example

$$\begin{aligned}
 R_{required} &= \frac{30}{\cos(70^\circ)} = 87.7 \text{ m} \\
 W_t &= 2(30) \tan(70^\circ) = 164.8 \text{ m} \\
 W_u &= 164.8 \text{ m} \\
 S &= 164.8(1 - 0.30) = 115.4 \text{ m} \\
 \Delta y &= \frac{164.8}{511} = 0.322 \text{ m} \\
 \Delta x &= \frac{6(0.514444)}{5} = 0.617 \text{ m} \\
 SPC &= \frac{4}{0.617(0.322)} = 20.1 \\
 N_{line} &= \left\lceil \frac{600}{115.4} \right\rceil + 1 = 7 \\
 L_{total} &= 7(1600) = 11200 \text{ m} \\
 N_{QC} &= \left\lceil \frac{11200(0.08)}{700} \right\rceil = 2 \\
 T_{total} &= 3.37 \text{ h}
 \end{aligned}$$

14.2 EM712 Medium-Depth Planning

14.2.1 Input Values

- Order 1b; Grid Size = 5 m; Required SPC = 2; Full Search = Off.
- Sonar Model = EM712; Planning Depth = 500 m; Frequency = 70 kHz.
- Swath Angle = 130°; Sonar Planning Range = 1000 m.
- Vessel Speed = 8 kn; Ping Rate = 2 Hz; Soundings per Ping = 512.
- Beam Mode = Equidistant; Overlap = 25%.

Range check

$$R_{required} = \frac{500}{\cos(65^\circ)} = 1183 \text{ m}$$

If $R_{sonar} = 1000 \text{ m}$, then:

$$R_{required} > R_{sonar}$$

The planner reduces the effective angle:

$$\alpha_r = \cos^{-1} \left(\frac{500}{1000} \right) = 60^\circ$$

$$\theta_e = 120^\circ$$

$$W_e = 2(500) \tan(60^\circ) = 1732 \text{ m}$$

14.3 EM124 Deep-Water Planning

14.3.1 Input Values

- Order 2; Grid Size = 10 m; Required SPC = 1; Full Search = Off.
- Sonar Model = EM124; Planning Depth = 4000 m; Frequency = 12 kHz.
- Swath Angle = 130°; Sonar Planning Range = 6000 m.
- Vessel Speed = 10 kn; Ping Rate = 0.5 Hz; Soundings per Ping = 512.
- Beam Mode = Equiangular; Overlap = 20%.

Range check

$$R_{required} = \frac{4000}{\cos(65^\circ)} = 9465 \text{ m}$$

If $R_{sonar} = 6000 \text{ m}$, then the requested swath angle cannot be achieved.

$$\alpha_r = \cos^{-1} \left(\frac{4000}{6000} \right) = 48.2^\circ$$

$$\theta_e = 96.4^\circ$$

$$W_e = 2(4000) \tan(48.2^\circ) = 8954 \text{ m}$$

15 Assumptions and Limitations

The planner uses simplified but transparent planning assumptions. It does not model every detail of real sonar systems or field operations.

15.1 General Limitations

- Constant depth, speed, sea state, and bottom return are assumed within the planning area.
- Dynamic beam steering, manufacturer-specific high-density modes, and bottom detection uncertainty are not modeled.
- Roll, pitch, heave, sound velocity refraction, and data cleaning losses are not directly simulated.
- Sonar profiles are planning profiles and not certified manufacturer specifications.
- Recommended QC Crossline percentages are planner defaults, not IHO S-44 prescribed percentages.

16 References

The final released manual should be reviewed against:

1. IHO S-44 Standards for Hydrographic Surveys.
2. Kongsberg EM2040 product documentation.
3. Kongsberg EM712 product documentation.
4. Kongsberg EM124 product documentation.

5. R2Sonic 2024/2026 product documentation.
6. Teledyne SeaBat T-series product documentation.
7. NOAA Hydrographic Surveys Specifications and Deliverables.
8. National hydrographic office survey instructions.
9. Project-specific survey specifications.